

Nimit Jain

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PROFESSIONAL EXPERIENCE

Kotak Mahindra Bank

Bengaluru, India

Data Scientist I, Commercial Banking

Apr 2026 – Present

- Drive the ongoing holistic collections turnaround strategy, cutting portfolio bounce rates by 4.3% in the initial 5 months to achieve the lowest monthly credit cost in 3 years while establishing sustained containment frameworks.
- Architected dynamic multi-tier telecalling allocation infrastructure routing pre-due and post-due accounts across internal agents, partner agencies, and automated voice bots, sharply curtailing early delinquency roll-forward rates.
- Engineered automated Apache Airflow pipelines tracking recovery KPIs across field agent hierarchies, accelerating MoM visit coverage to resolve unvisited account backlogs and lower portfolio-level roll-forward rates by 5.5%.
- Synthesized internal delinquency records with credit bureau datasets to detect off-us debt servicing leakage, deploying a targeted early repossession strategy that recovered Rs 5.8 Cr and boosted recovery efficiency by 23%.

Associate III, Commercial Banking

Jun 2025 – Mar 2026

- Scaled customer acquisition data pipelines in Python and SQL across 60M+ borrower records, applying bureau risk filters to size a Rs 9,000 Cr pre-approved loan segment that successfully steered Bharat Mortgages expansion.
- Designed an early-warning customer retention and churn engine for Agri-Business banking by tracking credit utilization drops and bureau inquiry surges, establishing proactive outreach workflows that cut client exits by 16% YoY.
- Automated PSI/CSI drift tracking and statistical hypothesis tests (t-tests, chi-square) across 7 production credit models, eliminating 28+ monthly hours of manual reporting while maintaining rigorous regulatory governance.
- Analyzed pin-code demographics, commercial market share, and branch serviceability radii to identify underserved commercial corridors, formulating data-backed geographic expansion roadmaps presented to CEO and MD.

STMicroelectronics

Greater Noida, India

Project Trainee, Embedded Systems and Power Electronics

Jul 2024 – Jun 2025

- Engineered and published the production-grade X-CUBE-ISO1 software package in C, implementing modular architectural patterns, strict dynamic memory management, unit test suites, and enterprise firmware release standards.
- Developed PyPWM, an object-oriented Python framework for algorithmic modeling, harmonic distortion numerical simulation, and interactive Matplotlib visual analytics of PWM inverter strategies, published at PCIM Asia 2025.

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani

Pilani, India

M.Sc. Mathematics and Bachelor in Engineering, Electrical and Electronics Engineering

2020 – 2025

- **Academic Profile:** CGPA: 7.48/10 | Merit-cum-Need Scholarship (Awarded and retained across all 10 academic semesters)
- **Relevant Coursework:** Probability & Statistics, Econometric Methods, Stochastic Processes, Mathematical Modeling, Numerical Optimization, Linear Algebra, Object-Oriented Programming, Data Structures & Algorithms, Real Analysis

PROJECTS

Autonomous Underwriting Agent & LLM Inference Engine

Personal Project

Tech Stack: PyTorch, vLLM, Hugging Face Transformers, ChromaDB (RAG), FastAPI, Docker

- Constructed an agentic retrieval-augmented generation (RAG) platform indexing historical commercial credit cases in ChromaDB to generate context-grounded, explainable underwriting rationales via locally hosted open LLMs.
- Optimized model serving using vLLM with continuous batching, PagedAttention, and KV caching, reducing p95 token generation latency by 42% while eliminating out-of-memory failures under high-concurrency production workloads.

Systematic Price Forecasting & Quantitative Risk Engine

Personal Project

Tech Stack: Python, XGBoost, Scikit-learn, Statsmodels, Pandas, Hypothesis Testing

- Built an end-to-end machine learning pipeline forecasting UK half-hourly electricity spot prices, engineering cyclical seasonality, extreme weather anomalies, and rolling autoregressive demand features across volatile regimes.
- Validated predictive performance via walk-forward cross-validation and residual normality tests, cutting RMSE by 18% over baseline models; backtested systematic execution strategies accounting for market transaction costs.

TECHNICAL SKILLS

- **Languages & Developer Tools:** Python, SQL, C, Java, MATLAB, Git, Apache Airflow, AWS (S3, Redshift, SageMaker)
- **Machine Learning & Generative AI:** PyTorch, vLLM, Hugging Face, Transformers, Scikit-learn, XGBoost, RAG, FastAPI
- **Mathematics & Analytics:** Hypothesis Testing, Regression Analysis, Time Series Forecasting, RCA, PSI/CSI Drift

PUBLICATIONS, ACHIEVEMENTS & LEADERSHIP

- **IEEE Xplore Publication:** Co-authored “PWM Strategies for VSIs: Python-Based Analysis, Simulation and Hardware Deployment”, PCIM Asia 2025; peer-reviewed technical conference publication indexed in IEEE Xplore digital library.
- **AlphaGrep Alphaverse Premiere League 2023:** Achieved Global Rank 4 of 1,400+ participants for developing quantitative algorithmic trading models and high-Sharpe systematic alpha strategies across simulated equity markets.
- **Vice President, Mathematics Association (BITS Pilani):** Led a 50-member student council organizing university academic symposia, quantitative workshops, and flagship national technical festival competitions for 1,500+ attendees.